

Test Design Techniques

SE401: Software Quality Assurance and Testing

Outline

- The test development process
- Categories of test design techniques
- Case Study – Knight Capital
- Specification based testing (Black box)
- Black Box Testing Techniques
- General Testing
- Experience based testing

Background

- Before we start testing, we need to know
 - What are we trying to test?
 - What are the inputs?
 - What are the results that should be produced by those inputs?
 - How do we prepare the tests?
 - How do we run the tests?
- To answer these questions we will look at
 - Test conditions
 - Test cases
 - Test procedures

Background

- The test design process can be done in different ways, from very informal (little or no documentation), to very formal.
- The level of formality depends on the context of the testing, including:



Test development process

1. Test analysis
2. Test design
3. Test implementation

1. Test analysis

- **The test basis documentation** is analyzed in order to determine *what* to test, i.e. to *identify the test conditions*.
- Test condition (*Def.*) = an item or event that could be verified by one or more test cases
- **Examples of test conditions**
 - A function
 - A transaction
 - A quality characteristic
 - Other structural elements (menus in web pages, etc.)

Test possibilities

- **“Throw a wide net!”**
- First; identify as many test conditions as possible
- Second; select which one to develop in more detail
- We can't test everything (P2). We have to select a subset of all possible tests, but this subset must have a high probability of finding most of the defects in the system.
- We need a suitable test design technique to guide our selection and to prioritize the test conditions.

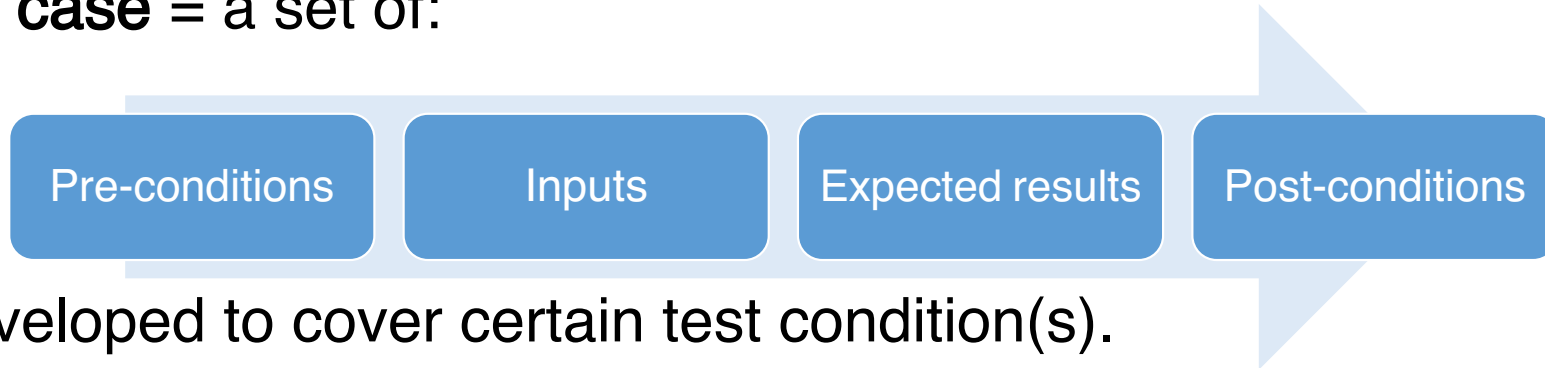
2. Test design

- During test design:



are created and specified.

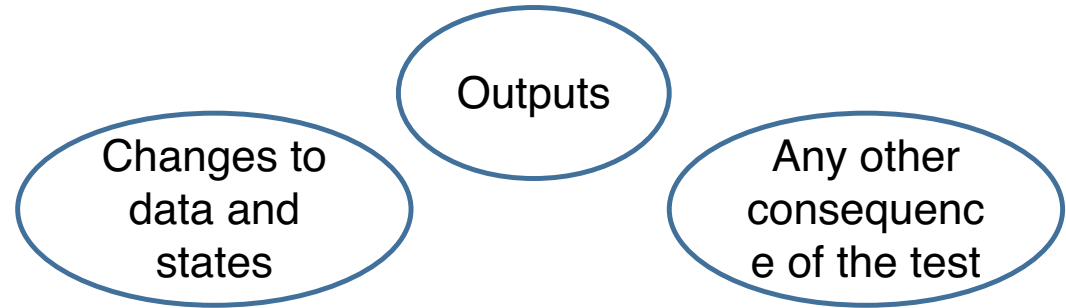
- **Test case** = a set of:



developed to cover certain test condition(s).

Test oracle

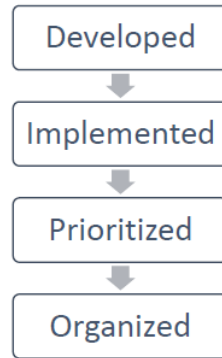
- In order to know what the system should do, we need to have a source of information about the correct behavior of the system an oracle
- Expected results include:



- If expected results have not been defined, then a plausible but erroneous result may be interpreted as the correct one
- Expected results should ideally be defined prior to test execution

3. Test implementation

- During test implementation the test cases are organized in the test procedures



- A manual test procedure
 - Specifies the sequence of action to be taken for executing of a test.
- An automated test procedure (test script)
 - If tests are run using a test execution tool , the sequence of action is specified in a test script

3. Test implementation

- The test execution schedule defines:
 - The order of execution of the test procedures & (possibly) automated test scripts
 - When they will be executed.
 - By whom to be executed.
- The test execution schedule will take into account such factors as:
 - risks
 - regression tests
 - prioritization
 - technical and logical dependencies

3. Test implementation

- Writing the test procedure is another opportunity to prioritize the tests, to ensure that the best testing is done in the time available.
- A good **rule of thumb** is 'Find the scary stuff first'. However the definition of what is 'scary' depends on the business, system or project and on the risks of the project.

Case Study – Knight Capital

High Frequency Trading (HFT)

Case Study – Knight Capital: High Frequency Trading (HFT)

- The ***Knight Capital Group*** is an American global financial services firm.
- Its high-frequency trading algorithms Knight was the largest trader in U.S. equities
 - with a market share of 17.3% on NYSE and 16.9% on NASDAQ.

Case Study – Knight Capital

- Aug 1, 2012.
 - In the first 45 minutes, Knight Capital's computers executed a series of unusually large automatic orders.
“... spit out duplicate buy and sell orders, jamming the market with high volumes of trades that caused the wild swings in stock prices.”
- By the end of day: \$460 million loss
 - *“Trading Program Ran Amok, With No ‘Off’ Switch”*
- In two days, the company's market value plunged by 75%



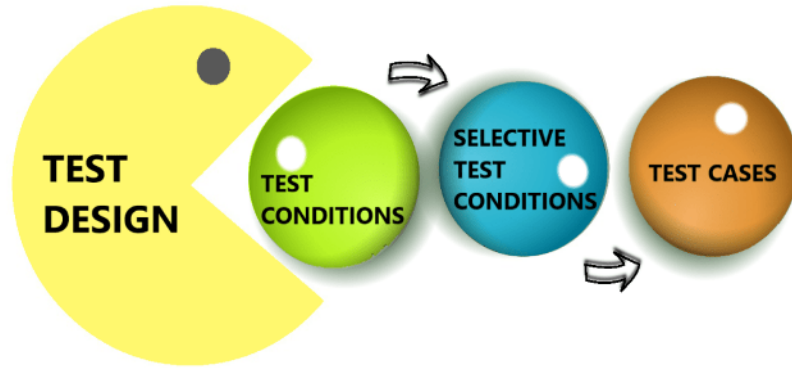
Case Study – Knight Capital: What Happened?

"Zombie Software" Blamed for Knight Capital Trading Snafu

- A new algorithmic trading program had just been installed, and began operation on Aug 1.
- A dormant legacy program was somehow "inadvertently reactivated"
- Once activated, the dormant system started multiplying stock trades by one thousand
 - Sent 4 million orders when attempting to fill just 212 customer orders
- “Knight’s staff looked through *eight* sets of software before determining what happened.”

Case Study – Knight Capital: The Investigation and Findings

- SEC launched an investigation in Nov 2012. Findings:
 - Code changes in 2005 introduced defects. Although the defective function was *not meant to be used*, it was kept in.
 - New code deployed in late July 2012. The defective function was triggered under new rules. Unable to recognize when orders have been filled.
 - Ignored system generated warning emails.
 - Inadequate controls and procedures for code deployment and testing.
- Charges filed in Oct 2013
 - Knight Capital settled charges for \$12 million



Testing Techniques

Testing techniques

Recall the difference between static and dynamic test techniques:

Static testing



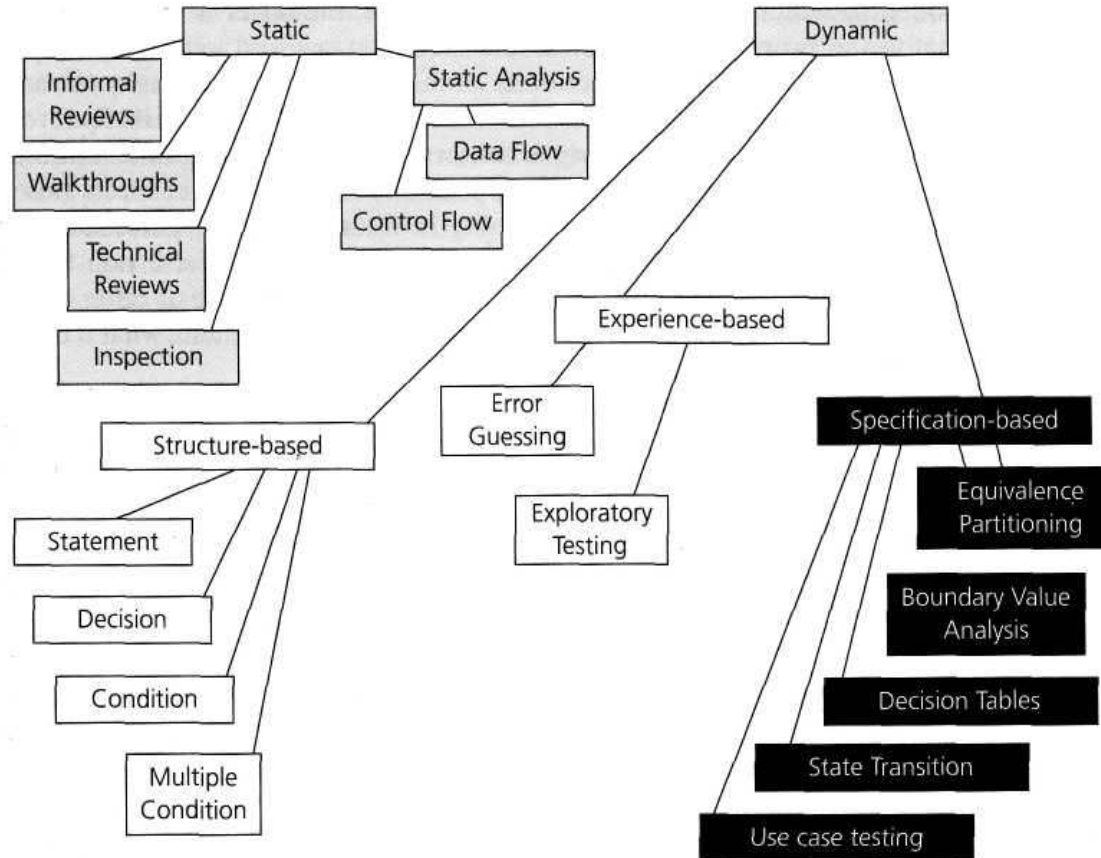
- Manual examination and automated analysis of code without executing it

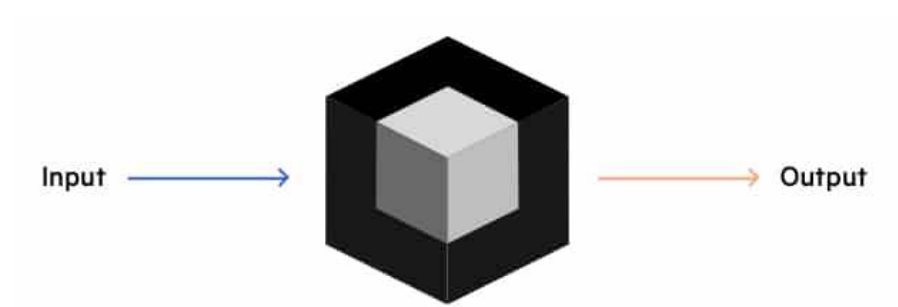
Dynamic testing



code to be executed

Categories of test design techniques





Specification-Based Testing

Black Box Testing

Functional Testing: A.k.a.: Black Box Testing

- Derive test cases from the *functional* specifications
 - *functional* refers to the source of information
 - not to what is tested
- *Also known as:*
 - specification-based testing (from specifications)
 - black-box testing (no view of the code)
- Functional specification = description of intended program behavior
 - either formal or informal

Common features of black box techniques

- Derive systematically test cases from specification
- Test cases are derived from how the software is constructed for example: code and design.
- Using existing test cases, we can measure the test coverage of the software
- Further test cases can be derived systematically to increase the test coverage
- The test cases are derived from the knowledge and experience of people:
 - Knowledge of testers, developers, users and other stakeholders about the software, its usage and its environment
 - Knowledge about likely defects and their distribution

Black Box Testing

- Testing software against a specification of its external behavior without knowledge of internal implementation details
 - Can be applied to software “units” (e.g., classes) or to entire programs
 - External behavior is defined in API docs, Functional specs, Requirements specs, etc.
- Because black box testing purposely disregards the program's control structure, attention is focused primarily on the information domain (i.e., data that goes in, data that comes out)
- The Goal: Derive sets of input conditions (test cases) that fully exercise the external functionality

Black-box Testing Errors Categories

- Incorrect or missing functions
- Interface errors
 - Usability problems
 - Concurrency and timing errors
- Errors in data structures or external data base access
- Behavior or performance errors
- Initialization and termination errors
- Unlike white box testing, black box testing tends to be applied later in the development process

Questions answered by Black-box Testing

- How is functional validity tested?
- How are system behavior and performance tested?
- What classes of input will make good test cases?
- Is the system particularly sensitive to certain input values?
- How are the boundary values of a data class isolated?
- What data rates and data volume can the system tolerate?
- What effect will specific combinations of data have on system operation?

The Information Domain: inputs and outputs

- Inputs

- Individual input values
 - Try many different values for each individual input
- Combinations of inputs
 - Individual inputs are not independent from each other
 - Programs process multiple input values together, not just one at a time
 - Try many different combinations of inputs in order to achieve good coverage of the input domain
- Ordering and Timing of inputs
 - In addition to the particular combination of input values chosen, the ordering and timing of the inputs can also make a difference

The Information Domain: inputs and outputs

- Defining the input domain
 - Boolean value
 - T or F
 - Numeric value in a particular range
 - $-99 \leq N \leq 99$
 - Integer, Floating point
 - Non-negative
 - One of a fixed set of enumerated values
 - {Jan, Feb, Mar, ...}
 - {Visa, MasterCard, Discover, ...}
 - Formatted strings
 - Phone numbers
 - File names
 - URLs
 - Credit card numbers
 - Regular expressions

Why Black Box Testing?

- Early.
 - can start *before* code is written
- Effective.
 - find some classes of defects, e.g., missing logic
- Widely applicable
 - any description of program behavior as spec
 - at any level of granularity, from module to system testing.
- Economical
 - less expensive than structural (white box) testing

The base-line technique for designing test cases

Early Black Box Testing

- Program code is not necessary
 - Only a description of intended behavior is needed
 - Even incomplete and informal specifications can be used
 - Although precise, complete specifications lead to better test suites
- Early test design has side benefits
 - Often reveals ambiguities and inconsistency in spec
 - Useful for assessing testability
 - And improving test schedule and budget by improving spec
 - Useful explanation of specification
 - or in the extreme case (as in XP), test cases are the spec

Functional versus Structural: Classes of faults

- Different testing strategies (functional, structural, fault-based, model-based) are most effective for different classes of faults
- Functional testing is best for *missing logic* faults
 - A common problem: Some program logic was simply forgotten
 - Structural (code-based) testing will never focus on code that isn't there!

Functional vs. Structural Test

- Functional test is applicable in testing at all granularity levels:
 - Unit test (from module interface spec)
 - Integration test (from API or subsystem spec)
 - System test (from system requirements spec)
 - Regression test (from system requirements + bug history)
- Structural test is applicable in testing relatively small parts of a system:
 - Unit test

Steps: From specification to test cases

1. Decompose the specification

- If the specification is large, break it into *independently testable features* to be considered in testing

2. Select representatives

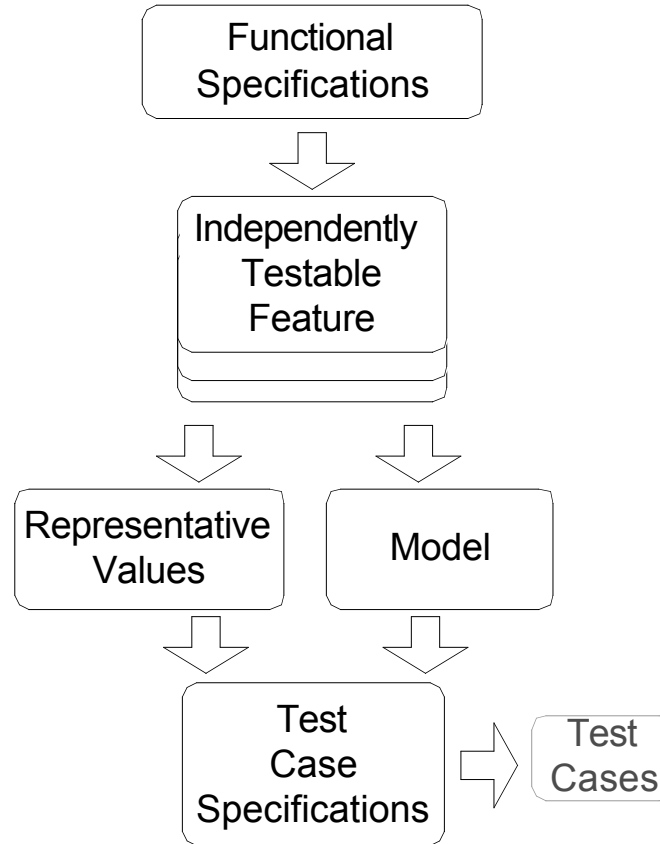
- Representative values of each input, or
- Representative behaviors of a *model*
 - Often simple input/output transformations don't describe a system. We use models in program specification, in program design, and in test design

3. Form test specifications

- Typically: combinations of input values, or model behaviors

4. Produce and execute actual tests

From specification to test cases



An Example: Postal Code Lookup



The screenshot shows the United States Postal Service logo at the top. Below it is a cartoon mailman holding a letter and a sign that says "ZIP CODE". To the right of the mailman is the text "ZIP Code Lookup". Below this are four buttons: "Search By Address >>", "Search By City >>", "Search By Company >>", and "Find". Below the buttons is the text "Find a list of cities that are in a ZIP Code." followed by a section titled "* Required Fields". Under this section is a label "* ZIP Code" next to an empty text input field. At the bottom of the form is a red "Submit >" button.

- Input: ZIP code (5-digit US Postal code)
- Output: List of cities

What are some representative values to test?

Example: Representative Values

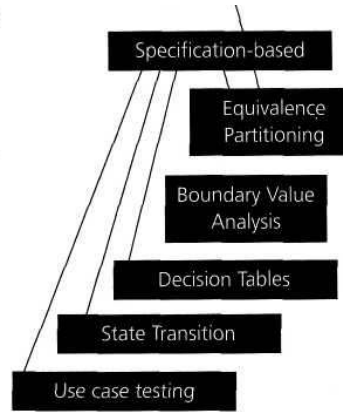
Simple example with
one input, one output

- Correct zip code
 - With 0, 1, or many cities
- Malformed zip code
 - Empty; 1-4 characters; 6 characters; very long
 - Non-digit characters
 - Non-character data

Note prevalence of boundary values (0 cities, 6 characters) and error cases



The screenshot shows the United States Postal Service ZIP Code Lookup interface. At the top is the USPS logo. Below it is a cartoon postman character holding a mailbag and a letter. The title "ZIP Code Lookup" is displayed. There are three tabs: "Search By Address >>", "Search By City >>", and "Search By Company >>". A "Find" button is on the right. Below the tabs, the instruction "Find a list of cities that are in a ZIP Code." is shown. Under "Required Fields", there is a label "* ZIP Code" followed by an empty text input field. At the bottom is a "Submit >" button.



Black Box Testing Techniques

Functional Testing Concepts

Single Defect Assumption: Failures are rarely the result of the simultaneous effects of two (or more) defects.

The four key concepts in functional testing are:

- Precisely identify the domain of each input and each output variable
- Select values from the data domain of each variable having important properties
- Consider combinations of special values from different input domains to design test cases
- Consider input values such that the program under test produces special values from the domains of the output variables

Equivalence Classes

- *The basic idea is to divide a set of test conditions into sub groups or sub sets (partitions) that can be considered the same .*
- *It is important that the different partitions do not have common elements*
- *We need only to test one condition from each partition , because all the conditions in the same partition will be treated in the same way by the software.*

Equivalence Partitioning

- A black-box testing method that divides the input domain of a program into classes of data from which test cases are derived
- An ideal test case single-handedly uncovers a complete class of errors, thereby reducing the total number of test cases that must be developed
- Test case design is based on an evaluation of equivalence classes for an input condition
- An equivalence class represents a set of valid or invalid states for input conditions
- From each equivalence class, test cases are selected so that the largest number of attributes of an equivalence class are exercised at once

Equivalence Partitioning

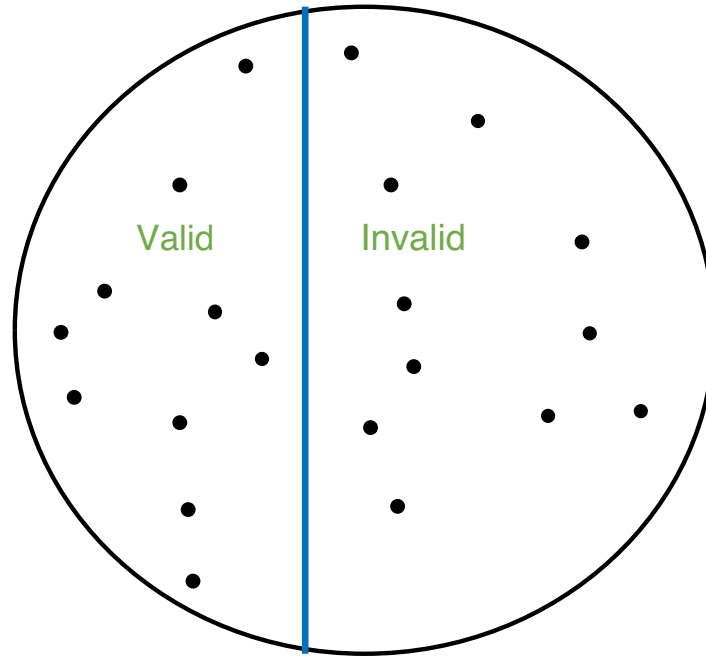
- Typically the universe of all possible test cases is so large that you cannot try them all
- You have to select a relatively small number of test cases to actually run
- Which test cases should you choose?
- Equivalence partitioning helps answer this question

Equivalence Partitioning

- Partition the test cases into "equivalence classes"
- Each equivalence class contains a set of "equivalent" test cases
- Two test cases are considered to be equivalent if we expect the program to process them both in the same way (i.e., follow the same path through the code)
- If you expect the program to process two test cases in the same way, only test one of them, thus reducing the number of test cases you have to run

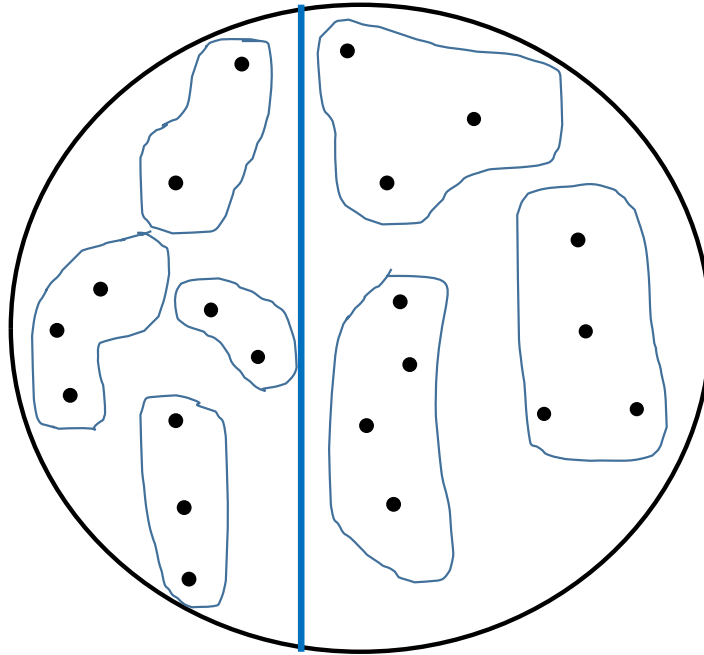
Equivalence Partitioning

- First-level partitioning: Valid vs. Invalid test cases



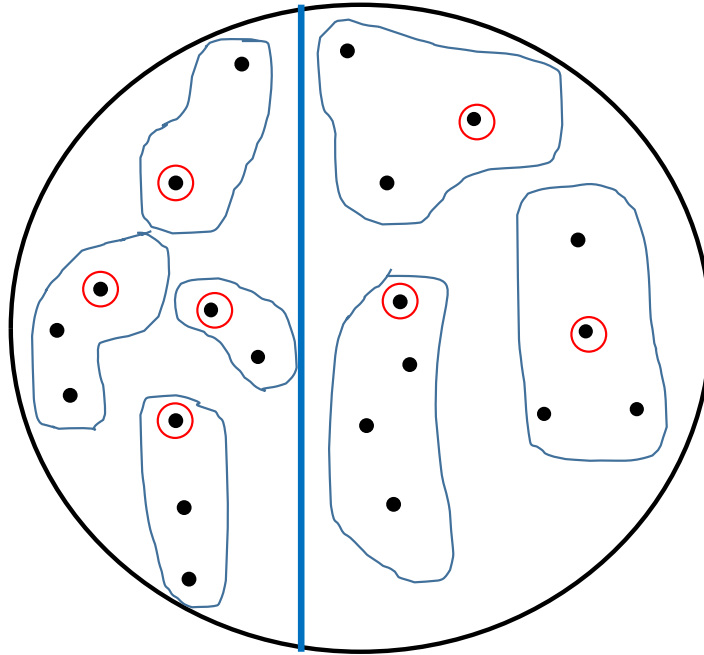
Equivalence Partitioning

- Partition valid and invalid test cases into equivalence classes



Equivalence Partitioning

- Create a test case for at least one value from each equivalence class



Equivalence Partitioning

- When designing test cases, you may use different definitions of “equivalence”, each of which will partition the test case space differently
 - Example: `int Add(n1, n2, n3, ...)`
 - Equivalence Definition 1: partition test cases by the number of inputs (1, 2, 3, etc.)
 - Equivalence Definition 2: partition test cases by the number signs they contain (positive, negative, both)
 - Equivalence Definition 3: partition test cases by the magnitude of operands (large numbers, small numbers, both)
 - Etc.

Equivalence Partitioning

- When designing test cases, you may use different definitions of “equivalence”, each of which will partition the test case space differently
 - Example: string Fetch(URL)
 - Equivalence Definition 1: partition test cases by URL protocol (“http”, “https”, “ftp”, “file”, etc.)
 - Equivalence Definition 2: partition test cases by type of file being retrieved (HTML, GIF, JPEG, Plain Text, etc.)
 - Equivalence Definition 3: partition test cases by length of URL (very short, short, medium, long, very long, etc.)
 - Same host
 - Etc.

Equivalence Partitioning

- Test multiple values in each equivalence class.
- Often you're not sure if you have defined the equivalence classes correctly or completely, and testing multiple values in each class is more thorough than relying on a single value.

Guidelines for Defining Equivalence Classes

- If an input condition specifies **a range**, one valid and two invalid equivalence classes are defined
 - Input range: 1 – 10 Eq classes: {1..10}, {x < 1}, {x > 10}
- If an input condition requires **a specific value**, one valid and two invalid equivalence classes are defined
 - Input value: 250 Eq classes: {250}, {x < 250}, {x > 250}
- If an input condition specifies **a member of a set**, one valid and one invalid equivalence class are defined
 - Input set: {-2.5, 7.3, 8.4} Eq classes: {-2.5, 7.3, 8.4}, {any other x}
- If an input condition is **a Boolean value**, one valid and one invalid class are defined
 - Input: {true condition} Eq classes: {true condition}, {false condition}

Determining Equivalence Classes

- Look for ranges of numbers or values
- Look for memberships in groups
- Some may be based on time
- Include invalid inputs
- Look for internal boundaries
- Don't worry if they overlap with each other —
 - better to be redundant than to miss something
- However, test cases will easily overlap with boundary value test cases

Equivalence Partitioning - examples

Input	Valid Equivalence Classes	Invalid Equivalence Classes
A integer N such that: -99 <= N <= 99	?	?
Phone Number Area code: [200, 999] Prefix: (200, 999] Suffix: Any 4 digits	?	?

Equivalence Partitioning - examples

Input	Valid Equivalence Classes	Invalid Equivalence Classes
A integer N such that: -99 <= N <= 99	[-99, -10] [-9, -1] 0 [1, 9] [10, 99]	?
Phone Number Area code: [200, 999] Prefix: (200, 999] Suffix: Any 4 digits	?	?

Equivalence Partitioning - examples

Input	Valid Equivalence Classes	Invalid Equivalence Classes
A integer N such that: -99 <= N <= 99	[-99, -10] [-9, -1] 0 [1, 9] [10, 99]	< -99 > 99 Malformed numbers {12-, 1-2-3, ...} Non-numeric strings {junk, 1E2, \$13} Empty value
Phone Number Area code: [200, 999] Prefix: (200, 999] Suffix: Any 4 digits	?	?

Equivalence Partitioning - examples

Input	Valid Equivalence Classes	Invalid Equivalence Classes
A integer N such that: $-99 \leq N \leq 99$	$[-99, -10]$ $[-9, -1]$ 0 $[1, 9]$ $[10, 99]$	< -99 > 99 Malformed numbers {12-, 1-2-3, ...} Non-numeric strings {junk, 1E2, \$13} Empty value
Phone Number Area code: [200, 999] Prefix: (200, 999] Suffix: Any 4 digits	555-5555 (555)555-5555 555-555-5555 $200 \leq \text{Area code} \leq 999$ $200 < \text{Prefix} \leq 999$	?

Equivalence Partitioning - examples

Input	Valid Equivalence Classes	Invalid Equivalence Classes
A integer N such that: $-99 \leq N \leq 99$	$[-99, -10]$ $[-9, -1]$ 0 $[1, 9]$ $[10, 99]$	< -99 > 99 Malformed numbers {12-, 1-2-3, ...} Non-numeric strings {junk, 1E2, \$13} Empty value
Phone Number Area code: [200, 999] Prefix: (200, 999] Suffix: Any 4 digits	555-5555 (555)555-5555 555-555-5555 $200 \leq \text{Area code} \leq 999$ $200 < \text{Prefix} \leq 999$	Invalid format 5555555, (555)(555)5555, etc.. Area code < 200 or > 999 Area code with non-numeric characters <i>Similar for Prefix and Suffix</i>

Equivalence Partitioning

- Technique
 - Inputs/outputs/internal values of the software are divided into groups that are expected to exhibit similar behavior .
- Examples
 - People (woman / man)
 - Students (bachelor / master)
 - Tickets (children / youth / adults /older)
 - Vehicles (gasoline / diesel / electric)

Equivalence Partitioning

- Technique
 - Inputs/outputs/internal values of the software are divided into groups that are expected to exhibit similar behavior .
- Equivalence partitions can be found for both valid data and invalid data, i.e. values that should be rejected.
- Notes
 - Tests can be designed to cover more than one partitions
 - Equivalence partitioning is applicable at all levels of testing.
 - Equivalence partitioning as a technique can be used to achieve input and output coverage

Boundary Value Testing

- Test values, sizes, or quantities near the design limits
 - value limits, length limits, volume limits
 - null strings vs. empty strings
- Errors tend to occur near the extreme values of inputs (off by one is an example)
- Robustness:
 - How does the software react when boundaries are exceeded?
- Use input values
 - at their minimum, just above the minimum
 - at a nominal value,
 - at the maximum, just below the maximum

Boundary Value Analysis

- A greater number of errors occur at the boundaries of the input domain rather than in the "center"
- Boundary value analysis is a test case design method that complements equivalence partitioning
 - It selects test cases at the edges of a class
 - It derives test cases from both the input domain and output domain

Guidelines for Boundary Value Analysis

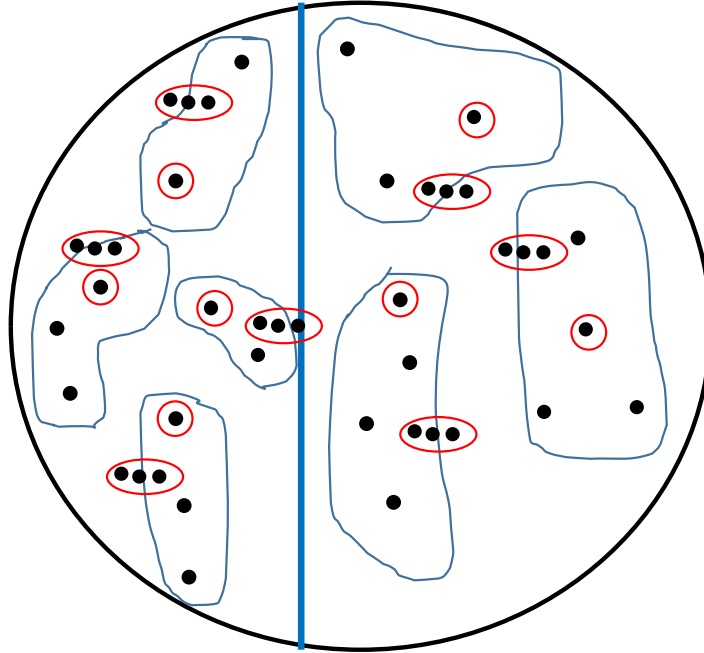
1. If an input condition specifies a range bounded by values a and b , test cases should be designed with values a and b as well as values just above and just below a and b
 2. If an input condition specifies a number of values, test case should be developed that exercise the minimum and maximum numbers. Values just above and just below the minimum and maximum are also tested
- Apply guidelines 1 and 2 to output conditions; produce output that reflects the minimum and the maximum values expected; also test the values just below and just above
 - If internal program data structures have prescribed boundaries (e.g., an array), design a test case to exercise the data structure at its minimum and maximum boundaries

Boundary Value Analysis

- When choosing values from an equivalence class to test, use the values that are most likely to cause the program to fail
- Errors tend to occur at the boundaries of equivalence classes rather than at the "center"
 - If `(200 < areaCode && areaCode < 999) { // valid area code }`
 - **Wrong!**
 - If `(200 <= areaCode && areaCode <= 999) { // valid area code }`
 - Testing area codes 200 and 999 would catch this error, but a center value like 770 would not
- In addition to testing center values, we should also test boundary values
 - Right on a boundary
 - Very close to a boundary on either side

Boundary Value Analysis

- Create test cases to test boundaries of equivalence classes



Boundary Value Analysis - examples

Input	Boundary Cases
A number N such that: -99 <= N <= 99	?
Phone Number Area code: [200 .. 999] Prefix: [200 .. 999] Suffix: Any 4 digits	?

Boundary Value Analysis - examples

Input	Boundary Cases
A number N such that: $-99 \leq N \leq 99$	-100, -99, -98 -10, -9 -1, 0, 1 9, 10 98, 99, 100
Phone Number Area code: [200 .. 999] Prefix: [200 .. 999] Suffix: Any 4 digits	?

Boundary Value Analysis - examples

Input	Boundary Cases
A number N such that: $-99 \leq N \leq 99$	-100, -99, -98 -10, -9 -1, 0, 1 9, 10 98, 99, 100
Phone Number Area code: [200 .. 999] Prefix: [200 .. 999] Suffix: Any 4 digits	Area code: 199, 200, 201 Area code: 998, 999, 1000 Prefix: 200, 199, 198 Prefix: 998, 999, 1000 Suffix: 3 digits, 5 digits

Boundary Value Analysis - examples

- Numeric values are often entered as strings which are then converted to numbers internally [int x = atoi(str);]
- This conversion requires the program to distinguish between digits and non-digits
- A boundary case to consider: Will the program accept / and : as digits?

Char	/	0	1	2	3	4	5	6	7	8	9	:
ASCII	47	48	49	50	51	52	53	54	55	56	57	58

Mainstream usage testing

- Don't get so wrapped up in testing boundary cases that you neglect to test "normal" input values
 - Values that users would typically enter during mainstream usage

Limitations of Boundary Value Testing

- Doesn't require much thought
- May miss internal boundaries
- Usually assumes the variables are independent
- Values at the boundary may not have meaning

Boundary value analysis

- Notes

- Boundary value analysis can be applied at all test levels .
- Its relatively easy to apply and its defect finding capability is high
- Detailed specifications are helpful.
- This technique is often considered as an extension of equivalence partitioning
- Boundary values are used for test data selection

Equivalence partitioning and boundary

- Why do both equivalence partitioning and boundary value analysis?
- Boundary values are usually extreme values
- To gain confidence to the system we also want to test it under normal circumstances
- Rule of thumb (Closed, valid partitions)
 - Pick two boundary values (min and max), and one value from the middle of the partition.

Guidelines and observations

- Equivalence Class Testing is appropriate when input data is defined in terms of intervals and sets of discrete values.
- Equivalence Class Testing is strengthened when combined with Boundary Value Testing
- Strong equivalence takes the presumption that variables are independent. If that is not the case, redundant test cases may be generated

Decision Table testing

- Decision tables are a good way
 - to capture system requirements that contain logical conditions
 - to document internal system design
 - to record complex business rules that a system is to implement
- Recall truth tables in mathematical logic :

	P	Q	$P \wedge Q$
Case 1	T	T	T
Case 2	T	F	F
Case 3	F	T	F
Case 4	F	F	F

Decision Table testing

- When creating decision tables, the specification is analyzed, and actions of the system are identified.
- If the input conditions and actions are stated in a way where they are either be true or false (Boolean), decision tables can be useful
- The decision table contains the triggering conditions
 - all combinations of true and false for all input conditions, and
 - the resulting actions for each combination of conditions.

Decision Table testing

- Decision tables
 - Cause-effect table
 - Used when input and actions can be expressed as Boolean
 - Systematic way of stating complex business rules
 - Help tester identify effects of combination of different input
 - Effective approach to reveal faults in the requirements

Conditions	R1	R2	R3	R4
Condition_1	T	T	F	F
Condition_2	T	F	T	F
Actions				
Action_1	?	?	?	?
Action_2	?	?	?	?

Decision Table testing

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Conditions								
<i>Condition 1</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Condition 2</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>
<i>Condition 3</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>
Actions								
<i>Action 1</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Action 2</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Action 3</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Action 4</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>

Decision Table testing

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Conditions								
<i>Condition 1</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Condition 2</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>
<i>Condition 3</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>
Actions	<i>Action 3</i>	x	<i>Action 1</i>	<i>Action 2</i>	<i>Action 4</i>	x	x	x

- The **coverage standard** commonly used with decision table testing is to have at least one test per column, which typically involves covering all combinations of triggering conditions.

Decision Table testing

	Rule 1	Rule 2 & 6	Rule 3	Rule 4	Rule 5	Rule 7 & 8
Conditions						
<i>Condition 1</i>	<i>True</i>	----	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>
<i>Condition 2</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>
<i>Condition 3</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	---
Actions	<i>Action 3</i>	x	<i>Action 1</i>	<i>Action 2</i>	<i>Action 4</i>	x

The **coverage standard** commonly used with decision table testing is to have at least one test per column, which typically involves covering all combinations of triggering conditions.

Decision Table testing

- **Example-**Decision table for credit-card

- If you are a new customer opening a credit card account, you will get a 15% discount on all your purchases today.
- If are an existing customer and you hold a loyalty card, you get 10% discount.
- If you have a coupon, you can get 20% off today (but it can't be used with a 'new-customer' discount)
- Discount are added, if applicable.

Decision Table testing

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Conditions								
<i>New customer (15%)</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Loyalty card (10%)</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>
<i>Coupon (20%)</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>
Actions								
<i>Discount</i>	x	x	20 %	15%	30%	10%	20%	0%

Decision Table testing

	Rule 1 & 2	Rule 3 & 7	Rule 4	Rule 5	Rule 6	Rule 8
Conditions						
<i>New customer (15%)</i>	<i>True</i>	<i>---</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>
<i>Loyalty card (10%)</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>
<i>Coupon (20%)</i>	<i>---</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>
Actions						
<i>Discount</i>	x	20 %	15%	30%	10%	0%

Decision Table testing

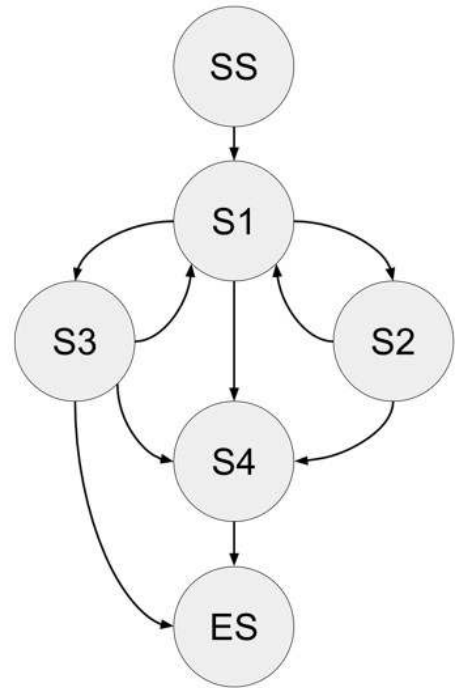
- **Decision tables** are a good way to:
 - capture system requirements that contain *logical conditions*
 - to document internal system design.
- The input conditions and actions are most often stated in such a way that they can be either true or false (Boolean).
- The strength of decision table testing is that it creates combinations of conditions that might not otherwise have been exercised during testing.
- It may be applied to all situations when the action of the software depends on several logical decisions.

State transition testing

- **A system** can be in a finite number of different states. This aspects of the system can be described as a *'finite state machine'*; a *state diagram*.
- Any system where you get a different output for the same input, depending on what has happened before, is a finite state system.
- The transition from one state to another are determined by the rules of the 'machine'.

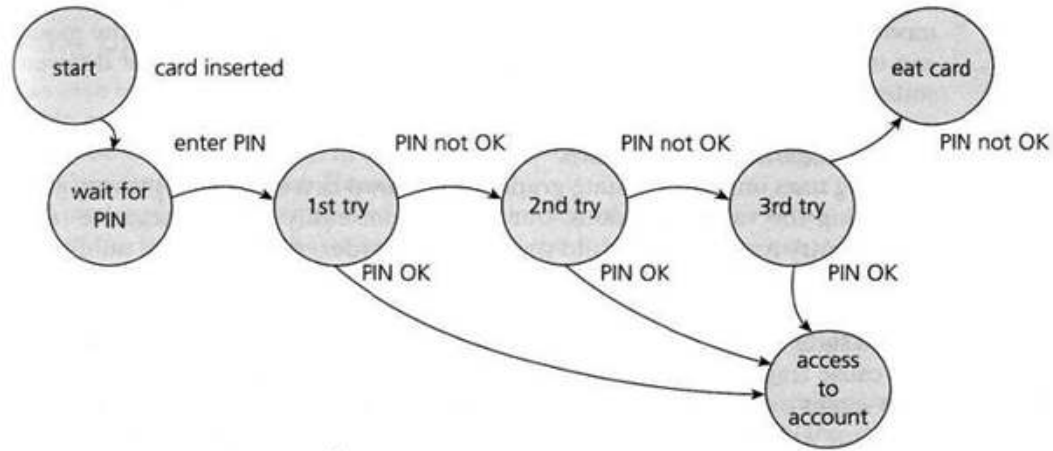
State transition testing

- State transition testing
 - System can be in a finite number of different states
- Elements of state transition models
 - States (The software may occupy)
 - Open/closed, active/inactive
 - Transitions (From one state to another)
 - Not all transitions are allowed
 - Events (Causing state transition)
 - Closing a files/withdrawing money
 - Actions (Action resulting from transitions)
 - Error message



State transition testing

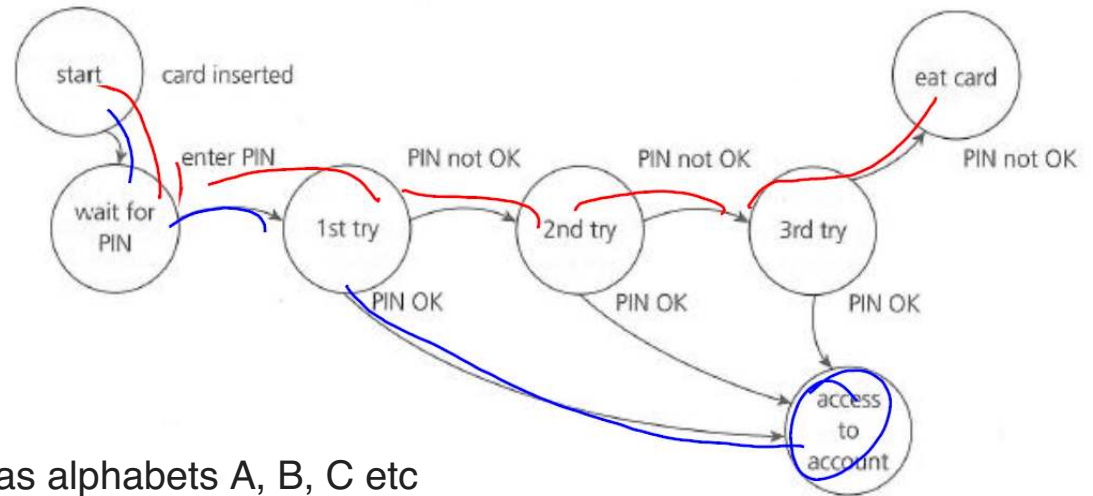
- A 'finite state machine' is often shown as a state diagram
- ATM PIN example.



- The states of the system under test are separate, identifiable and finite in number.

State transition testing

- How many tests do we need to exercise every state?



States: States can be numbered as S1, S2 or as alphabets A, B, C etc

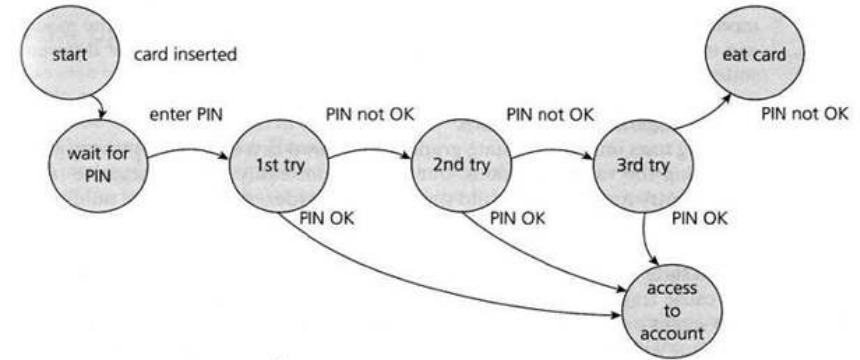
S1:Start, S2:Wait for Pin, S3: 1st try, S4: 2nd Try, S5: 3rd Try, S6: access to account, S7: eat card

Events:

Event1:Card inserted, Event 2: enter Pin, Event 3: Pin OK, Event 4: Pin not OK

Actions : (not shown in the above example) could be : Messages on the screen – error or otherwise.

State transition testing



	Event 1 (Insert card)	Event 2 (Enter Pin)	Event 3 (Pin OK)	Event 4 (Pin not OK)
S1:Start	S2	-	-	-
S2:Wait for Pin	-	S3	-	-
S3: 1st try	-	-	S6	S4
S4: 2nd Try	-	-	S6	S5
S5: 3rd Try	-	-	S6	S7
S6: access to account	-	-	-	-
S7: eat card	S1	-	-	-

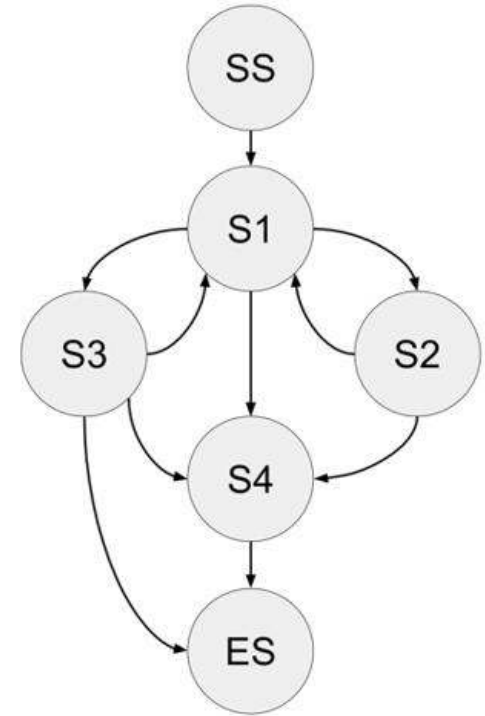
- Invalid or Null Transitions are represented as '-' in red in the table above.

State transition testing

- Why state transition testing?
 - Because a system may exhibit a different response depending on current conditions or previous history.
- State transition testing allows the tester to view:
 - the software in terms of its states
 - transitions between states
 - the inputs or events that trigger state changes (transitions)
 - the actions which may result from those transitions

State transition testing

- Tests can be designed
 - to cover a typical sequence of states
 - to exercise specific sequences of transitions
 - to cover every state
 - to exercise every transition
 - to test invalid transitions
- State transition testing is much used within the software industry and technical automation in general.



Use case testing

- Use case describes interactions between actors (users and the system), which produce a result of value to a system user

Example

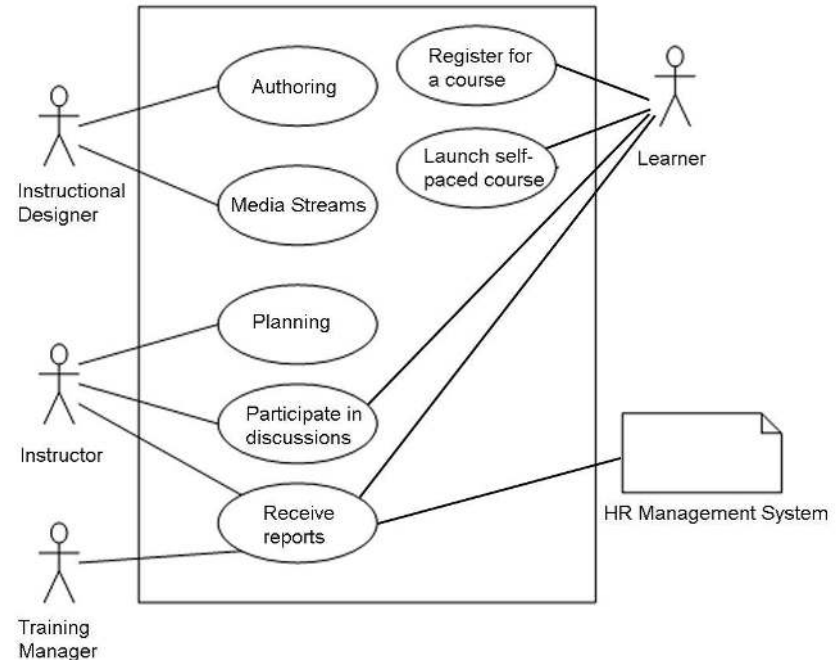
An on-line training website:

User 1: the learner

User 2: the tutor (instructor)

User 3: the training manager

User 4: the instructional designer



Use case testing

- Identify test cases that exercise the whole system on a transaction by transaction basis from start to finish.
- Describe interactions between actor and system
- Use the language and terms of the business rather than technical terms, especially when the actor is a business user.
- Can uncover integration defects, that is, defects caused by the incorrect interaction between different components.

Use case name	<name>	
Actor(s)	<actor1>, ...	
Pre-conditions	<cond1>, ...	
Post-conditions	<cond1>, ...	
Main Success Scenario	Step	Description
	1	A: <action>
	2	S: <response>
	3	A: <action>
	4	S: <response>
	...	...
Extensions	Step	Description
	S.X	<cause> S: <response>
	S.Y	<cause> S: <response>

Use case testing

- Each use case has pre-conditions, which need to be met for a use case to work successfully.
- Each use case terminates with post-conditions, which are the observable results and final state of the system after the use case has been completed.
- A use case usually has a *mainstream*(i.e. most likely) scenario, and sometimes *alternative branches*.

Use case testing

Main Success Scenario A: Actor S: System	Step	Description
	1	A: Inserts card
	2	S: Validates card and asks for PIN
	3	A: Enters PIN
	4	S: Validates PIN
	5	S: Allows access to account
Extensions	2a	Card not valid S: Display message and reject card
	4a	PIN not valid S: Display message and ask for re-try (twice)
	4b	PIN invalid 3 times S: Eat card and exit

Use case testing

- Very useful for designing acceptance tests with customer/user participation.
- Describe the 'process flows' through a system base on its actual likely use.
- Derived from use cases are most useful in uncovering defects in the process flows during real-world use of the system.
- Help uncover integration defects caused by the integration and interference of different components, which individual testing would not see.
- Designing test cases from use cases may be combined with other specification-based test techniques.

Summary: Key Concepts

- Black-box testing
 - vs. random testing, white-box testing
- Black box testing techniques
 - Equivalence class
 - Boundary value testing
 - Decision tables
 - State transition
 - Use case testing

General Testing

Testing for race conditions and other timing dependencies

- Many systems perform multiple concurrent activities
 - Operating systems manage concurrent programs, interrupts, etc.
 - Servers service many clients simultaneously
 - Applications let users perform multiple concurrent actions
- Test a variety of different concurrency scenarios, focusing on activities that are likely to share resources (and therefore conflict)
- "Race conditions" are bugs that occur only when concurrent activities interleave in particular ways, thus making them difficult to reproduce
- Test on hardware of various speeds to ensure that your system works well on both slower and faster machines

Performance Testing

- Measure the system's performance
 - Running times of various tasks
 - Memory usage, including memory leaks
 - Network usage (Does it consume too much bandwidth? Does it open too many connections?)
 - Disk usage (Is the disk footprint reasonable? Does it clean up temporary files properly?)
 - Process/thread priorities (Does it play well with other applications, or does it hog the whole machine?)

Performance Testing

- Tools
 - IBM Rational Performance Tester
 - Apache Jmeter
 - WebLOAD
 - LoadRunner

Limit Testing

- Test the system at the limits of normal use
- Test every limit on the program's behavior defined in the requirements
 - Maximum number of concurrent users or connections
 - Maximum number of open files
 - Maximum request size
 - Maximum file size
 - Etc.
- What happens when you go slightly beyond the specified limits?
 - Does the system's performance degrade dramatically, or gracefully?

Stress Testing

- Test the system under extreme conditions (i.e., beyond the limits of normal use)
- Create test cases that demand resources in abnormal quantity, frequency, or volume
 - Low memory conditions
 - Disk faults (read/write failures, full disk, file corruption, etc.)
 - Network faults
 - Unusually high number of requests
 - Unusually large requests or files
 - Unusually high data rates (what happens if the network suddenly becomes ten times faster?)
- Even if the system doesn't need to work in such extreme conditions, stress testing is an excellent way to find bugs

Security Testing

- Any system that manages sensitive information or performs sensitive functions may become a target for intrusion (i.e., hackers)
- How feasible is it to break into the system?
- Learn the techniques used by hackers
- Try whatever attacks you can think of
- Hire a security expert to break into the system
- If somebody broke in, what damage could they do?
- If an authorized user became disgruntled, what damage could they do?

Security Testing

- Tools:
 - Metasploit
 - W3af
 - Zed Attack Proxy (ZAP)
 - Wapiti

Usability Testing

- Is the user interface intuitive, easy to use, organized, logical?
- Does it frustrate users?
- Are common tasks simple to do?
- Does it conform to platform-specific conventions?
- Get real users to sit down and use the software to perform some tasks
- Watch them performing the tasks, noting things that seem to give them trouble
- Get their feedback on the user interface and any suggested improvements
- Report bugs for any problems encountered

Usability Testing

- Tools:
 - Crazy Egg
 - Optimizely
 - Usabilla

Recovery Testing

- Try turning the power off or otherwise crashing the program at arbitrary points during its execution
 - Does the program come back up correctly when you restart it?
 - Was the program's persistent data corrupted (files, databases, etc.)?
 - Was the extent of user data loss within acceptable limits?
- Can the program recover if its configuration files have been corrupted or deleted?
- What about hardware failures? Does the system need to keep working when its hardware fails? If so, verify that it does so.

Configuration Testing

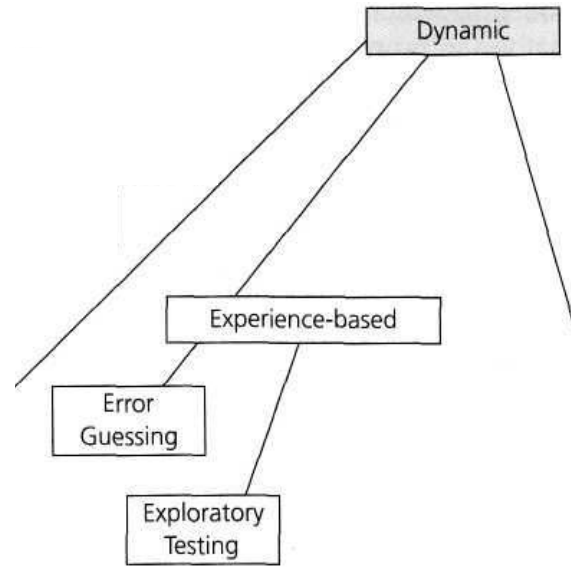
- Test on all required hardware configurations
 - CPU, memory, disk, graphics card, network card, etc.
- Test on all required operating systems and versions thereof
 - Virtualization technologies such as VMWare and Virtual PC are very helpful for this
- Test as many Hardware/OS combinations as you can
- Test installation programs and procedures on all relevant configurations

Compatibility Testing

- Test to make sure the program is compatible with other programs it is supposed to work with
 - Ex: Can Word 12.0 load files created with Word 11.0?
 - Ex: "Save As... Word, Word Perfect, PDF, HTML, Plain Text"
 - Ex: "This program is compatible with Internet Explorer and Firefox"
- Test all compatibility requirements

Documentation Testing

- Test all instructions given in the documentation to ensure their completeness and accuracy
- For example, “How To ...” instructions are sometimes not updated to reflect changes in the user interface
- Test user documentation on real users to ensure it is clear and complete



Experience-based techniques

Experience-based techniques

- Tests are derived from the tester's skill and intuition and their experience with similar applications and technologies.
- When used to augment systematic techniques, experienced based testing can be useful in identifying special tests not easily captured by formal techniques, especially when applied after more formal approaches.
- May yield widely varying degrees of effectiveness , depending on the testers experience.

Experience-based techniques

- Error guessing = a commonly used experienced-based technique.
- Generally testers anticipate defects based on experience.
- A structured approach to the error guessing technique is to enumerate a list of possible errors and to design tests that attack these errors.
- This systematic approach is called fault attack.

Experience-based techniques

- Exploratory testing = concurrent test design, test execution, test logging and learning,
- *based on a test charter containing test objectives, and carried out within time-boxes.*
- It is most useful ...
 - where there are few or inadequate specifications
 - under severe time pressure
 - to complement other, more formal testing
 - It can serve to help ensure that the most serious defects are found.

Special Value Testing

- The most widely practiced form of functional testing
- The tester uses his or her domain knowledge, experience, or intuition to probe areas of probable errors
- Other terms: “hacking”, “out-of-box testing”, “ad hoc testing”, “seat of the pants testing”, “guerilla testing”

Uses of Special Value Testing

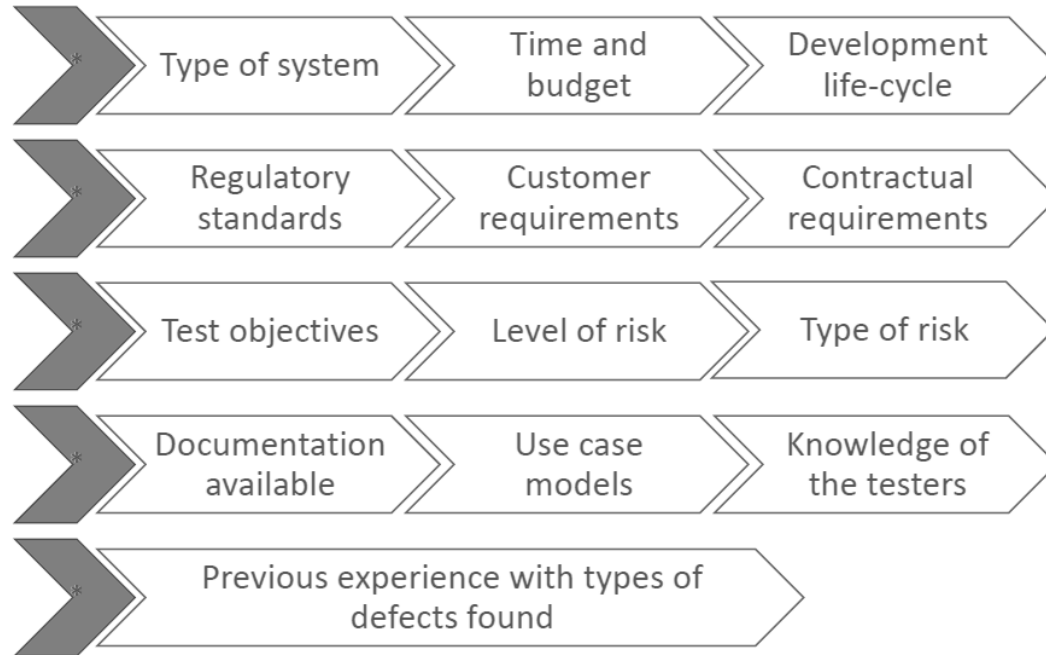
- Complex mathematical (or algorithmic) calculations
- Worst case situations (similar to robustness)
- Problematic situations from past experience
- “Second guess” the likely implementation

Characteristics of Special Value Testing

- Experience really helps
- Frequently done by the customer or user
- Defies measurement
- Highly intuitive
- Seldom repeatable
- Often, very effective

Choosing test techniques

- The choice of which test techniques to use depends on a number of factors, including:



Summary

- Case Study – Knight Capital
- Testing techniques
- Black Box Testing
- Black Box Testing Techniques
- General Testing
- Experience based testing